

SO3-Cell: Standard Cell Layout Automation Framework for Simultaneous Optimization of Topology, Placement, and Routing

Chung-Kuan Cheng¹, Andrew B. Kahng¹, Byeonggon Kang^{1*}, Seokhyeong Kang², Jakang Lee², Bill Lin¹

¹University of California San Diego, USA ²POSTECH, Republic of Korea

{ckcheng, abk, b8kang, billin}@ucsd.edu, {shkang, wkrkd95}@postech.ac.kr

Abstract—We propose SO3-Cell, the first automatic standard cell layout generation framework that optimizes three key steps simultaneously using Mixed-Integer Linear Programming (MILP). SO3-Cell simultaneously performs circuit topology optimization, transistor placement, and internal cell routing to achieve an optimized layout solution. Our optimization objective is to minimize metal usage while enhancing cell layout flexibility within a given area.

We introduce design space pruning techniques to mitigate the complexity of larger designs, such as a full adder, a reset flip-flop (FF), and a 2-bit FF. We successfully generate a layout for a 44-transistor 2-bit FF within 25,862 seconds, demonstrating the scalability and robustness of the SO3-Cell framework. We evaluate the block-level PPA impact of the proposed cell-layout improvements, demonstrating a 35.0% reduction in power, a 2.2% increase in frequency, and a 31.1% reduction in area.

Index Terms—standard cell library, mixed-integer linear programming, pin accessibility, layout automation, placement, routing

I. INTRODUCTION

Automatic generation of standard cell layout is a critical component of modern VLSI design. This enables Design-Technology Co-Optimization (DTCO) to sustain Moore's law [1]. A standard cell layout generator facilitates the comparison of different technology options by generating diverse layouts, supporting DTCO efforts for next-generation technologies [2]. Furthermore, by formulating the optimization problem using Satisfiability Modulo Theories (SMT) or Integer Linear Programming (ILP), and solving it with state-of-the-art solvers, cell layout automation can efficiently explore the design space and generate optimized layout solutions. Ultimately, this helps in meeting the chip's Power, Performance, and Area (PPA) targets. The primary objective of a standard cell layout automation is to find an optimal solution for given technology constraints (e.g., Design Rules (DRs)) while maintaining a reasonable runtime. However, achieving layout optimality within a feasible runtime presents a fundamental trade-off. Because the problem is NP-hard [3], optimality in the worst case necessitates an exhaustive search of the design space, thereby resulting in significantly increased runtime. Cheng *et al.* [4] review and compare the optimality achieved by prior standard cell layout generators and further discuss the types of solvers employed to address the associated optimization problems.

Existing layout synthesis automation flows [5]–[10] suffer from three major limitations. First, most cell layout generators operate with a given circuit topology, limiting optimality to the circuit level. Figure 1(a) shows that topology optimization can reduce metal usage in the AOI21_X1 layout by swapping the transistor stack order. Figure 1(b) illustrates the area difference in the XOR2_X1 layout resulting from variations in the series stack order of the circuit. These examples highlight the critical relationship between circuit topology and layout optimality.

Second, heuristic placement algorithms reduce design space for speed but sacrifice optimality, leading to suboptimal layouts. Pre-

* Corresponding author.

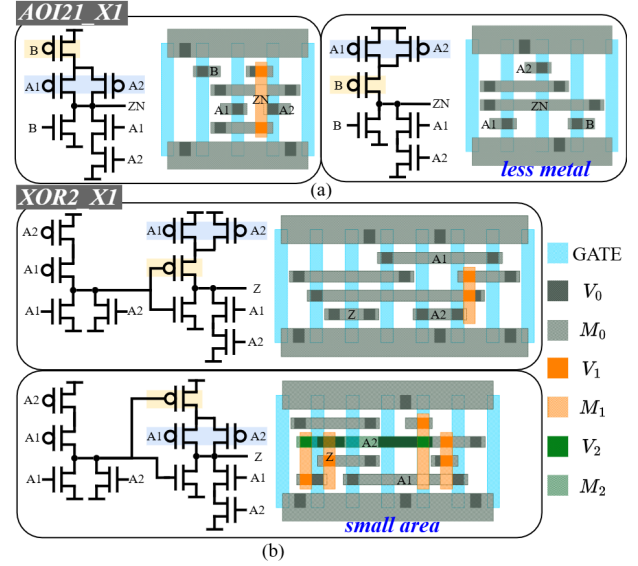


Fig. 1. Circuit topology influences layout quality. (a) shows the difference in metal usage for AOI21_X1, while (b) illustrates the area variation in XOR2_X1, both depending on the circuit topology.

processing transistor pairing [6] of n PMOS and n NMOS transistors reduces the placement search space from $O(n! \cdot 2^{2n})$ to $O(n! \cdot 2^{2n})^1$, and can guarantee optimality for small combinational cells. However, in cases such as flip-flops (FFs), where multiple transistors share the same gate input net, or when PMOS and NMOS transistors placed in the same column are required to use different signals (i.e., when the gate nets differ between the paired NMOS and PMOS transistors) to achieve an optimal solution, this condition does not hold. Transistor partitioning can further reduce runtime for larger cells by constraining placement, but since the relative locations of transistors are not known in advance, it may depend on predefined solutions. Moreover, using more than two partitions can degrade optimality, as it introduces additional complexity in determining the placement of each partition.

Third, successful routing at the cell level requires consideration of multiple physical factors due to increasing design complexity and tighter integration. As Moore's Law progresses, standard cell layouts shrink, reducing the number of available routing tracks. However, the number of pins per cell remains constant, increasing pin density and making routing more challenging. While traditional approaches primarily focus on pin access points [11], advanced technology nodes

¹Without pairing, each PMOS and NMOS transistor can be placed independently with flipping, resulting in $(n! \cdot 2^n)^2$ combinations. With pre-pairing, only $n!$ pair arrangements are needed, and each pair can still flip independently, yielding $n! \cdot 2^{2n}$ combinations.

demand more comprehensive metrics [12]—such as routing scenarios, placement flexibility, and track utilization—to enhance chip-level PPA. Kang *et al.* [12] introduce the Cell-Flex metrics, which define *cell layout flexibility* as the extent to which a cell contributes to the quality of block-level implementation. These metrics consist of Pin Access Flexibility (PAF), Cell Placement Flexibility (CPF), and Track Utilization Flexibility (TUF). These must be formulated as constraints and integrated into cell layout automation to enhance block-level PPA.

This paper proposes SO3-Cell, the first standard cell generator that optimizes three key steps simultaneously using Mixed-Integer Linear Programming (MILP). SO3-Cell integrates circuit topology optimization, transistor placement, and internal cell routing to achieve an optimized layout solution. We define optimality through a scalarized objective that minimizes total metal usage (dominant term) and maximizes *cell layout flexibility* within a given area. Our contributions in this work are summarized as follows:

- 1) **Integrating circuit topology optimization into cell layout generation.** We propose SO3-Cell, the first automatic cell layout generation framework that simultaneously performs circuit topology optimization, transistor placement, and internal cell routing to generate an optimized layout.
- 2) **Introducing design space pruning techniques.** We develop efficient design space pruning strategies that preserve optimality, including symmetry breaking, maximum occupying track, misaligned gate specification, and structure-driven partitioning.
- 3) **Incorporating Cell-Flex metrics into layout generation.** We generate layout patterns optimized for Cell-Flex metrics while also minimizing metal usage.
- 4) **Validating the impact of SO3-Cell through block experiments.** We demonstrate the effectiveness of the SO3-Cell-generated libraries through block-level experiments, showing improvements in PPA.

II. SO3-CELL: SIMULTANEOUS OPTIMIZATION OF TOPOLOGY, PLACEMENT AND ROUTING

In this section, we describe the SO3-Cell framework. Subsection II-A provides an overview of SO3-Cell, Subsection II-B presents complexity-aware optimization strategies, and Subsection II-C presents the preliminary settings underlying our framework.

A. Overview of SO3-Cell

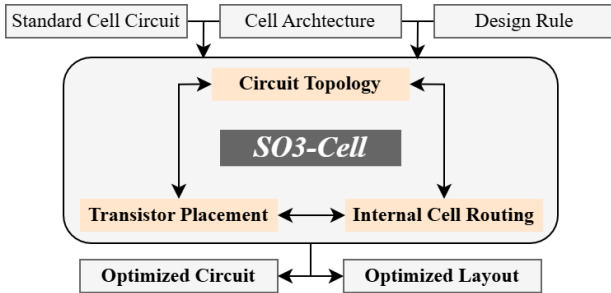


Fig. 2. Overall flow of our SO3-Cell framework.

SO3-Cell simultaneously performs circuit topology optimization, transistor placement, and internal cell routing. The dynamic formulation of all three steps and the incorporation of Cell-Flex metric [12] formulations enable the generation of optimized circuit and layout solutions. Figure 2 presents an overview of the SO3-Cell framework. The inputs are standard cell circuits described in netlist format, cell architecture (e.g., metal track configuration, cell height, and other

physical constraints), and DR information. Our studies are based on the PROBE3.0 Process Design Kit (PDK) [13]. The key features for cell library generation are presented in Table I and the notations are shown in Table II.

TABLE I
KEY FEATURES OF OUR CELL LIBRARY GENERATION. THE ACRONYMS EOL, MAR, AND OVL STAND FOR END OF LINE RULE, MINIMUM AREA RULE, AND OVERLAP RULE (ALSO KNOWN AS THE VIA ENCLOSURE RULE), RESPECTIVELY.

Cell Architecture					
Cell Height			Single (SH), Double (DH)		
# of M_0 & M_2 Track			4 & 4		
Contacted Poly Pitch (CPP)			45 nm		
M_1 Pitch			30 nm		
Design Rules					
Metal Layer			Via Layer		
	EOL	MAR		OVL (lower)	OVL (upper)
M_0	10 nm	483 nm ²	V_0	5 nm	5 nm
M_1	38 nm	1080 nm ²	V_1	5 nm	5 nm
M_2	38 nm	756 nm ²	V_2	5 nm	5 nm

TABLE II
NOTATIONS FOR THE SO3-CELL FRAMEWORK.

Term	Description
M_x, V_x	The x^{th} metal and via layers
IO	External I/O pins
N	Set of nets n
$a(v)$	Set of vertices adjacent to v
$e_{u,v,r}$	The r^{th} edge between u and v , $u \in a(v)$
$e_{u,v,r}^n$	0-1 indicator if $e_{u,v,r}$ used by net n
$w_{u,v,r}$	Weighted cost assigned to edge $e_{u,v,r}$
S_i, P_i	The i^{th} series/parallel node of tree
T_i	Set of transistors contained in S_i, P_i
$G_{i,j,k}$	The k^{th} edge between node i and j
t_q	The q^{th} transistor in T_i
$o(c)$	Number of occupied tracks in column c

B. Complexity-Aware Optimization Strategy (CA-OptS)

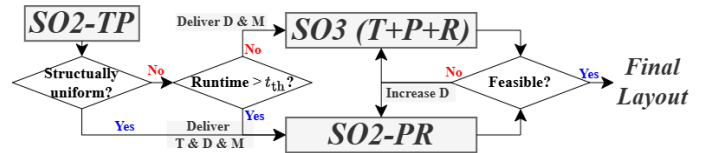


Fig. 3. Complexity-Aware Optimization Strategy (CA-OptS) for scalable layout generation. T is circuit Topology, P is transistor Placement, R is internal cell Routing, D is # of Dummy gates and M is # of Misaligned gates.

We propose a complexity-aware optimization strategy (CA-OptS) to reduce the complexity of the SO3-Cell operation. Figure 3 shows the CA-OptS flow. **SO2-TP** is a preprocessing stage that co-optimizes circuit topology and transistor placement while determining two complexity indicators—the number of dummy transistors D and the number of misaligned gates M —and infers the minimum cell size implied by (D, M) . Based on this initial assessment, the flow branches into two modes: **SO2-PR** if the circuit is structurally uniform (one diffusion network with identical transistors per stack) or if the SO2-TP runtime exceeds t_{th} ; otherwise, **SO3**, a unified formulation that jointly optimizes topology, placement, and internal routing under a single objective. CA-OptS provides both flexibility and scalability, enabling efficient exploration of complex designs while maintaining layout quality.

The objective function of the SO3-Cell aims to minimize the overall metal usage while maximizing *cell layout flexibility* in the given D and M . The objective is defined as follows:

$$\text{Minimize : } \sum_{e_{u,v,r} \in E} w_{u,v,r} \cdot e_{u,v,r} - (\text{PAF} + \text{TUF})$$

The objective function consists of two components: the total metal usage and a reward term based on Cell-Flex metrics. The first term, $w_{u,v,r} \cdot e_{u,v,r}$, represents the total metal length, computed as the weighted sum of routing edges. The second term, $\text{PAF} + \text{TUF}$, serves as a reward to improve Cell-Flex. CPF is reflected in the first term of the objective function through $w_{u,v,r}$, whose details are provided in Section V-B.

C. Preliminary Settings

Figure 4 illustrates the four-layer 3D routing grid of our simultaneous layout framework. In the figure:

- **1st layer (Placement and Topology):** This layer represents the enumeration of circuit topologies and their corresponding transistor placements and net assignments on the grid. Based on the selected configuration in this layer, routing exploration is performed across the subsequent layers (2nd layer to 4th layer).
- **2nd layer (Classification):** This layer classifies the net placement results from the 1st layer by checking the alignment within the same column. This classification simplifies the number of active vertices for the routing.
- **3rd and 4th layers (Routing):** These layers represent the horizontal metal layers used for routing. Specifically, the 3rd layer models the first horizontal metal layer (M_0) and the 4th layer models the second horizontal metal layer (M_2). The edges between these two layers represent vias and vertical metal segments on M_1 .

In advanced technologies with unidirectional routing and track counts of 6 or fewer, only one vertical metal segment on M_1 can be used per column to connect M_0 to either another M_0 or an M_2 segment [14]. Leveraging this, we collapse the three-layer sequence— V_1 (connecting M_0 to M_1), vertical routing on M_1 , and V_2 (connecting M_1 to M_2)—into a *compound via edge* between the 3rd and 4th layers. This edge jointly encodes via usage (V_1 , V_2) and vertical M_1 occupancy. Specifically, (i) if the *compound via edge* is selected and incident to both an M_0 edge and an M_2 edge, then the connection traverses $M_0 \xrightarrow{V_1} M_1 \xrightarrow{V_2} M_2$; (ii) if the *compound via edge* is selected but there is no incident M_2 edge, it denotes a connection only between M_0 and M_1 (i.e., V_1 without V_2).

For double-height cells, we introduce an *inter-row edge* connecting the 4th layer in the lower and upper rows. This edge represents M_1 segments that bridge the two rows to provide inter-row connections. Inter-row connections are allowed only in columns that use the *compound via edge*. (i) If the *inter-row edge* is used, it implies that exactly one vertical M_1 segment bridges the lower and upper rows. (ii) If the *inter-row edge* is not used, it implies that any M_1 usage is confined to a single row (lower or upper). The added *inter-row edge* eliminates explicit M_1 modeling even in double-height cells.

This 3D routing grid is optimized for unidirectional routing in advanced technology nodes. Each vertex in the 3rd and 4th layers is incident to multiple routing edges, denoted as $e_{u,v,r}$, where u and v are adjacent vertices and r indicates the routing track index. In contrast, edges between the 2nd–3rd and 3rd–4th layers are single edges for layer transitions.

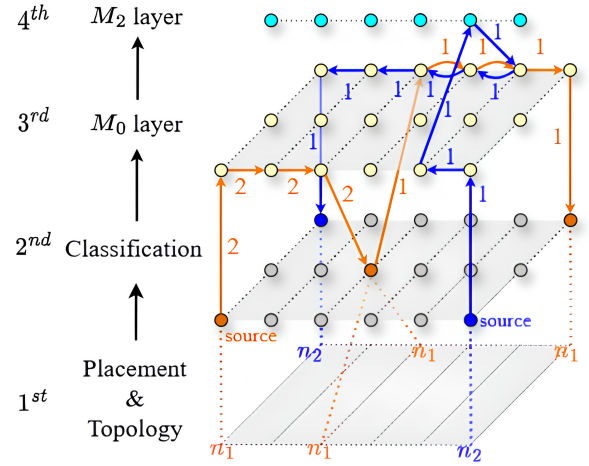


Fig. 4. Four-layer 3D routing grid used in the SO3-Cell framework. The example illustrates the routing of nets n_1 and n_2 through the placement and routing layers. The underlying flow network formulation follows the methodology of [15].

For feasible routing, we do not allow the routing edge $e_{u,v,r}$ to be shared by multiple nets. This enforces via exclusivity by ensuring that each routing edge is assigned to at most one net:

$$e_{u,v,r}^{n_1} + e_{u,v,r}^{n_2} \leq 1 \quad \forall n_1 \neq n_2 \in N, \quad \forall (u, v, r)$$

Our routing flow follows the methodology proposed in [15]. To reduce runtime during routing, we apply a bounding box constraint introduced in [16], implemented within an ILP formulation.

III. CIRCUIT TOPOLOGY OPTIMIZATION

We integrate circuit topology optimization into SO3-Cell, which explores the entire design space of functionally equivalent circuit topologies to identify the one that yields the optimized layout. Subsection III-A introduces the hierarchical series-parallel tree structure of the given circuit. Based on this structure, Subsection III-B presents the constraints for topology enumeration, and Subsection III-C describes the hierarchical net remapping for preserving circuit functionality.

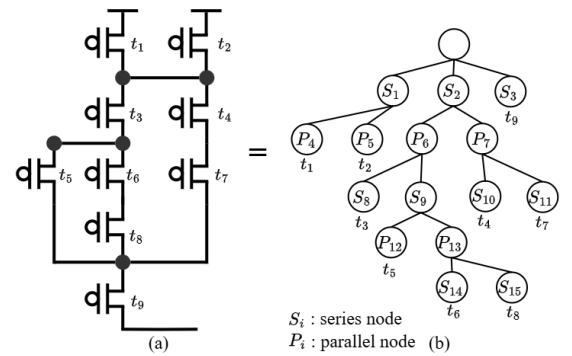


Fig. 5. Transformation of the circuit in (a) into a hierarchical series-parallel tree structure, shown in (b).

A. Hierarchical Series-Parallel Tree Structure

Figure 5(b) shows the series-parallel graph derived from the circuit in (a). In this multiple-level hierarchy, series (S) and parallel (P) nodes appear repeatedly as the tree expands. Each edge stores net information about the source and drain (S/D). Each leaf node corresponds to a single transistor. This transformed structure allows us to

analyze the connectivity and relative locations among the transistors. Such hierarchical decomposition enables adding the formulation of constraints between transistors while preserving the functionality of the overall circuit.

B. Series-Parallel Topology Enumeration

We enumerate all potential circuit topologies by swapping the edge connections of the child nodes. Let T_i denote the set of transistors contained in the i^{th} node, and let $G_{i,j,k}[t_q]$ represent a binary variable that indicates whether a specific transistor $t_q \in T_i$ is assigned under the k^{th} edge $G_{i,j,k}$, which connects child node i to its parent node j .

To enable topology-level swapping, we define the following constraints for each node with k edges. These are recursively applied at every level of the hierarchical structure.

- **Parallel constraint:** All transistors belonging to parallel nodes under the same parent node are assigned to the same edge.

$$G_{i,j,k}[t_q] - G_{i,j,k}[t_r] = 0, \quad \forall t_q, t_r \in T_i, t_q \neq t_r, \forall k$$

- **Series constraint:** All transistors belonging to different series nodes under the same parent node must not be assigned to the same edge.

$$G_{i,j,k}[t_q] + G_{i,j,k}[t_r] \leq 1, \quad \forall t_q \in T_{i_q}, t_r \in T_{i_r}, i_q \neq i_r, \forall k$$

- **Edge assignment constraint:** All transistors belonging to each node must be assigned to exactly one edge at their level.

$$\sum_k G_{i,j,k}[t] = 1, \quad \forall t$$

C. Hierarchical Net Remapping

Since swapping nodes changes the transistor's original S/D-net assignments, net alignment is required. To enable this, we introduce a hierarchical net remapping that systematically propagates a net name to maintain the functionality. Figure 6 illustrates net propagation across

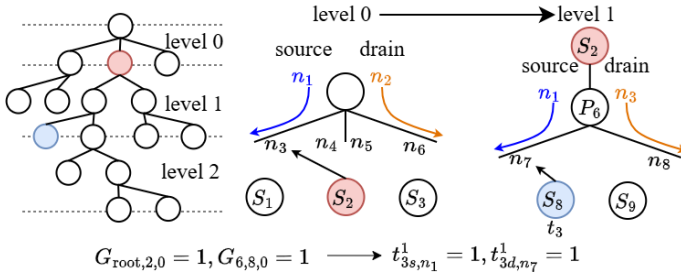


Fig. 6. Example illustrating net assignment variables t_{3s}^1 and t_{3d}^1 of transistor t_3 in node S_8 , based on edge selections G in both S_2 and S_8 .

levels, where net information is assigned to edges. The S/D-nets of transistor t_3 , located in node S_8 at level 1, are determined by the edge selection from its grandparent node S_2 . For each S/D-net candidate n at levels $l-1$ and l , we define corresponding binary variables $t_{3s,n}^l$ and $t_{3d,n}^l$ at level l . The following constraints define the source remapping for transistor t_3 in Figure 6.

- **Top-Down Net Propagation:** The net assignment propagates from level $l-1$ to level l , and the same logic applies to the drain variable $t_{3d,n}^l$.

$$t_{3s,n}^l \leq t_{3s,n}^{l-1}, \quad \forall n \in \{n_1, n_4, n_6\}$$

- **Edge-Dependent Selection (from upper level):** Source nets propagated from level $l-1$ are activated only if the first edge

($i = 0$) is selected, while drain nets are activated only if the last edge is selected.

$$\sum_{n \in \{n_1, n_4, n_6\}} t_{3s,n}^{l-1} = G_{6,8,0}[t_3]$$

- **Edge-to-Net Binding (from current level):** If an edge other than $i = 0$ is selected, its corresponding net is assigned as the source. For drain nets, the same logic applies when the last edge is selected.

$$\sum_{n \in \{n_8\}} t_{3s,n}^l = 1 - G_{6,8,0}[t_3], \quad t_{3s,n_8}^l \leq G_{6,8,1}[t_3]$$

By recursively applying these constraints to both S/D-nets of every transistor in the tree, we determine S/D-nets top-down, ensuring functional correctness throughout the hierarchy.

IV. DESIGN SPACE PRUNING TECHNIQUES

In this section, we introduce a design space pruning technique to mitigate the complexity of large cells. While the SO3 framework can generate optimized layouts for most combinational cells, applying full three-step simultaneous optimization becomes impractical for larger designs due to the exponential runtime increase caused by exhaustive design space exploration. Through the CA-OptS introduced in Subsection II-B and the techniques described in Subsections IV-A to IV-D, our framework significantly reduces runtime while preserving layout quality.

A. Symmetry Breaking

Figure 7 illustrates the symmetry-breaking methodology that we

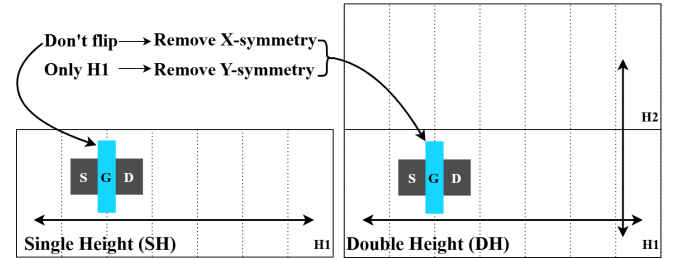


Fig. 7. Reducing the design space by symmetry breaking. For Single-Height (SH) cells, restricting the flipping of a single transistor eliminates duplicated X-symmetry cases. For Double-Height (DH) cells, further restricting placement to height-1(H1) removes both X- and Y-symmetry cases.

implement. Single-height cells exhibit X-symmetry, while double-height cells exhibit both X- and Y-symmetry. Due to this symmetry, each layout can generate duplicate optimized solutions. In single-height cells, we can eliminate this redundancy by restricting the flipping of a single transistor. In double-height cells, we can further reduce Y-symmetry by restricting the same transistor to be placed only at Height 1, H1.

B. Maximum Occupying Track

$$o_{s/d}^p(c) + o_{s/d}^n(c) + o_M(c) + o_C(c) \leq T_{\max}(M_0) + T_{\max}(M_2)$$

Figure 8 and the above formulation illustrate how we exclude infeasible placement cases by enforcing constraints based on the maximum allowable track usage $T_{\max}$ in M_0 and M_2 . This constraint accounts for track usage due to vias ($o_{s/d}^{\{p,n\}}$, o_M), shown in Figure 8(a), and track usage due to net crossings (o_C), shown in Figure 8(b). For every S/D-net column $c-1$ and $c+1$ adjacent to a gate column c , if the same net appears in any column other than c and its neighbor, a via must

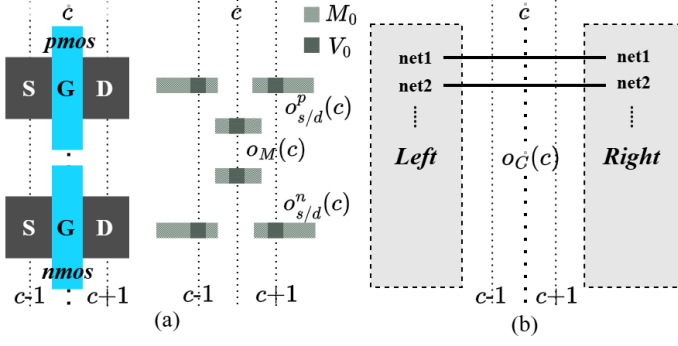


Fig. 8. (a) $o_{s/d}^{\{p,n\}}$ refers to the track occupancy due to vias used for S/D-net connections. o_M represents the track occupancy caused by misaligned gates. If a gate is misaligned, two via pickups are required, occupying two tracks. (b) Track occupancy $o_C(c)$ caused by net crossings between the left and right of $c-1$ and $c+1$.

be inserted in $c-1$ or $c+1$. Since a via occupies one routing track and columns $c-1$ and $c+1$ can share a track, a binary indicator $o_{s/d}^{\{p,n\}}(c)$ is sufficient to represent whether the track is used. To pick up the gate signal at column c , one via is needed per gate, or two vias if PMOS and NMOS transistors use different signals. Additionally, $o_C(c)$ captures the track usage caused by net crossings between the left and right of column c , with each shared net requiring at least one horizontal track.

By summing these components, we can estimate the minimum number of horizontal tracks required at column c and evaluate routing feasibility in advance.

C. Misaligned Gate Specification

If the nets connected to the gates of the PMOS and NMOS are different, each of the PMOS and NMOS requires a via for gate connection, resulting in the occupation of two tracks. However, misaligned gate columns are unavoidable to shrink cell size, particularly in complex cells such as FFs. Through the SO2-TP, we obtain the number of dummy transistors (D) and the number of misaligned gates (M). At this stage, the specific net names responsible for gate misalignment can be identified. We significantly reduce the design space by limiting misalignment to only the identified nets, effectively pruning unnecessary combinatorial exploration of misalignment configurations.

D. Structure-Driven Partitioning

For optimality and a bottom-up approach, we do not apply partitioning techniques, except for the 2-bit FF, which inherently consists of two identical FFs sharing the same structure and controlled by a common clock signal from a shared clock buffer.

TABLE III

Comparison of 2-bit FF with different partitioning strategies (P_1 , P_2 and P_3) to a 1-bit FF. Shared* refers to the clock buffer within a 2-bit FF, which is commonly used by both individual FFs. X indicates no partition constraint.

We aim to place the shared logic freely to find the optimized solution.

Type	Height	Partition Strategy			#CPPs	# M_1	# M_2
		1bit	2bit	Shared*			
1-bit	SH	X	N/A	X	14	3	1
1-bit	DH	X	N/A	X	14	4	0
2-bit P_1	SH	Left	Right	X	25	8	2
2-bit P_2	DH	Left	Right	X	26	6	0
2-bit P_3	DH	Top	Bottom	X	26	2	0

Table III presents our partitioning strategy for the 2-bit FF and compares CPP, which represents the cell size, as well as the number

of M_1 and M_2 layers. The 2-bit P_1 achieves the best area reduction; however, due to horizontal track limitations, it results in increased metal usage. The 2-bit P_3 , which follows a double-height (DH) architecture, utilizes only two M_1 layers and achieves an area reduction ratio of 0.93 compared to two 1-bit FFs ($26\text{CPP}/14\text{CPP} \times 2$). Figure 9 illustrates the layouts of P_1 and P_3 .

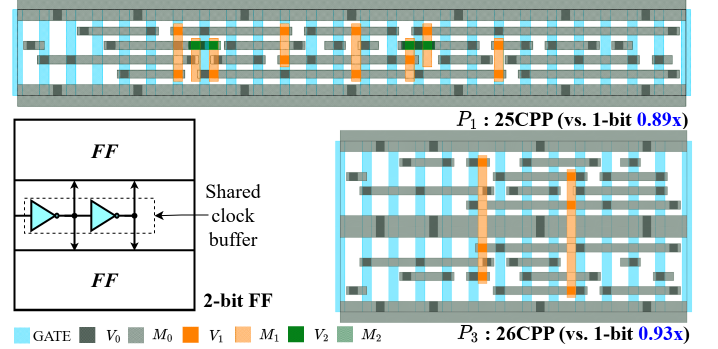


Fig. 9. 2-bit FF layouts generated using partitioning strategies P_1 and P_3 . While P_1 achieves a smaller area, P_3 requires only two M_1 tracks for routing.

V. INCORPORATING CELL-FLEX INTO SO3-CELL

By incorporating Cell-Flex metrics [12] into our framework, we enhance layout patterns in a way that promotes high-quality block-level PPA. Subsections V-A to V-C introduce the formulation for enhancing Cell-Flex and highlight its impact on layout changes.

A. Pin Stretchability and Separation for PAF

To increase the number of feasible routing scenarios within a cell, while considering pin interruptions and metal obstacles (OBS), we define two key aspects: **pin stretchability** ($\mathcal{P}_h$) (horizontal) and **pin separation** ($\mathcal{P}_v$) (vertical).

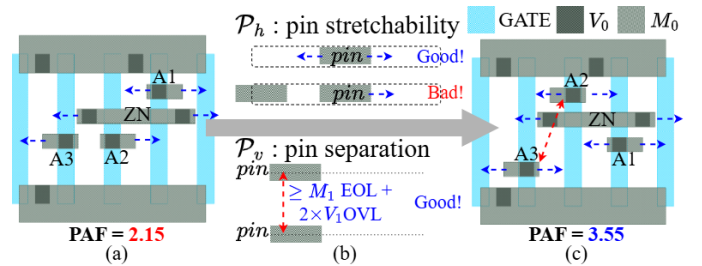


Fig. 10. NAND3_X1 layout improvement using pin stretchability $\mathcal{P}_h$ and pin separation $\mathcal{P}_v$. (a) The baseline layout without rewards results in a PAF of 2.15. (b) Visualization of the reward concepts. (c) Enhanced layout with applied rewards achieves a PAF of 3.55.

Figure 10 illustrates how we define $\mathcal{P}_h$ and $\mathcal{P}_v$ by using an enhanced NAND3_X1 layout as an example. A $\mathcal{P}_h$ reward is assigned when the pin metal segment is the only metal segment within the $\mathcal{P}_h$ checking boundary. This condition ensures horizontal stretchability without interference from other metals. The corresponding constraint is defined as:

$$\text{length of } \mathcal{W}_r \times (1 - \mathcal{P}_h) \geq \sum_{(u,v,r) \in \mathcal{W}_r} \left(\sum_{n \in N} e_{u,v,r}^n - e_{u,v,r}^{\text{pin}} \right)$$

$$1 - \mathcal{P}_h \leq \sum_{(u,v,r) \in \mathcal{W}_r} \left(\sum_{n \in N} e_{u,v,r}^n - e_{u,v,r}^{\text{pin}} \right)$$

where $\mathcal{W}_r$ denotes a window of five consecutive vertex pairs on the same routing layer, serving as the $\mathcal{P}_h$ checking boundary. This region is illustrated by the dotted line in Figure 10(b). This equation verifies that only one pin metal occupies the region, qualifying it for a $\mathcal{P}_h$ reward.

If adjacent pins are vertically separated by a distance greater than or equal to the sum of the upper layer's EOL spacing and twice the OVL, they can both be routed using the same metal track without violating DRs. This vertical separation reduces pin-access conflicts during detailed routing.

B. PDN-Aware Guidance for CPF

A stable power supply is essential for modern VLSI design to ensure a sufficient power integrity margin. This stability is achieved through a uniformly distributed power delivery network (PDN) [17]. However, the global metal tracks occupied by PDN metal constrain the placement locations of cells. To address this issue, SO3-Cell assigns a higher cost ($w_{u,v,V} + \Delta$) to via edges V that align with the PDN locations. This avoids placement restrictions caused by overlaps between PDN metal and internal routing. A high-weighted cost applied to tracks that may overlap with the PDN transforms the layout of the reset-FF in Figure 11 from the upper to the lower configuration, resulting in an improved CPF.

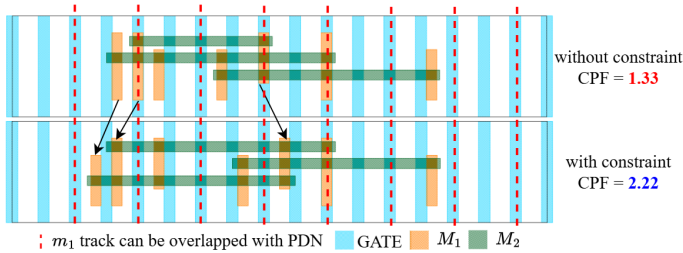


Fig. 11. Reset-FF layout enhancement. CPF of the reset-FF increases from 1.33 to 2.11.

C. OBS Alignment for TUF

The objective function of our framework is to minimize metal usage,

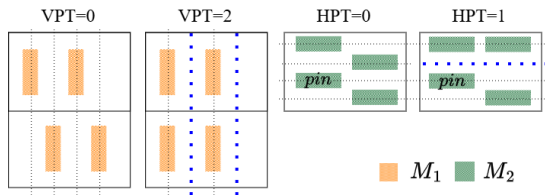


Fig. 12. Aligning the OBS enhances both vertical passing tracks (VPT) and horizontal passing tracks (HPT).

which improves TUF. However, a key challenge is preserving as many passing tracks as possible. Figure 12 illustrates how the metal arrangement can be improved through constraints. In the left plot, four M_1 OBS segments occupy different tracks, resulting in zero vertical passing tracks (VPT). By aligning these OBS segments using the proposed formulation, VPT can be maximized. A similar effect is shown in the right plot for horizontal passing tracks (HPT). These improvements help reduce detours in block-level routing, contributing to overall routing efficiency.

Let $OBS_t^{M_x} \in \{0, 1\}$ be a binary variable indicating whether track t on metal layer M_x is occupied by OBS. For each M_x track t , we define:

$$OBS_t^{M_x} \leq \sum_{n \in \mathcal{N}} \sum_{e_{u,v,r} \in t} e_{u,v,r}^n$$

$$OBS_t^{M_x} \geq e_{u,v,r}^n \quad \forall n \notin \text{IO}, \quad \forall e_{u,v,r}^n \in t$$

VI. EXPERIMENTS

We implement the SO3-Cell (Simultaneous Optimization of Three Steps–Cell) [19] framework using *Gurobi Optimizer v12.0.1* via its *Python* API, and execute it on a Linux machine equipped with an Intel Xeon E5-2620 CPU (2.40 GHz) and 256 GB of memory. Our implementation includes a netlist parser, allowing the framework to take a standard cell netlist as input. The output consists of text-based layout information, including transistor placement (i.e., location of placed transistors), net routing (i.e., target nets for in-cell routing), occupied track positions, and associated metal and via layer details. We validate DRC based on the DRs summarized in Table I, and use the PROBE3.0 PDK [13] to perform LVS and PEX using *Cadence Pegasus v21.3* and standard cell characterization at the (TT, 0.7 V, 25°C) corner. Liberty characterization is conducted using *Cadence Liberate v23.1*, following the Non-Linear Delay Model (NLDM). We use *Synopsys Fusion Compiler v22.12* (FC) for synthesis and place-and-route (P&R). FC facilitates block-level comparison of standard cell libraries by simultaneously performing placement and logic synthesis with advanced optimization features.

A. Library Configuration and Cell Generation

Table IV summarizes the cell generation results using the SO3-Cell framework, including cell specifications, evaluation metrics, and generation outcomes. The table focuses on cells newly added or enhanced compared to PROBE3.0. Since AOI and OAI cells share the same layout pattern when Y-flipped, they are grouped in a single row. The same approach is applied to XOR2 and XNOR2. The #FETs column demonstrates that our framework successfully scales to support up to 44 transistors, highlighting its scalability and robustness. The Height column indicates whether the DH architecture is applied; we apply DH to complex cells (e.g., FFs, full adder (FA_X1)) to improve layout quality. Cell evaluation metrics include the #CPPs, which reflect cell size, the number of metal layers used (#M1 and #M2), and the Cell-Flex metrics [12]. The Cell-Flex metrics comprise PAF, CPF, and TUF; the TUF metric decomposes into VPT, HPT, Vertical-to-Horizontal Via Availability (V2HVA), and Horizontal-to-Vertical Via Availability (H2VVA). The applied optimization strategy (OPT) is determined as follows: SO2-PR is applied if the circuit is structurally uniform (e.g., AOI22, NAND2) or if the SO2-TP runtime exceeds a threshold $t_{th} = 30$ sec; otherwise, SO3 is used. The presence of a newly discovered circuit topology is marked in the NCT column, and the runtime in seconds is reported in the Time column.

Table V compares the Cell-Flex metrics of cells generated by SO3 with those generated by PROBE3.0. The values are normalized as (SO3 – PROBE3.0), and for all metrics except Area, higher values indicate better performance. The results demonstrate that SO3 achieves superior metrics in most cases compared to PROBE3.0.

B. Effectiveness of Design Space Pruning

Table VI presents the runtime comparison for complex cells when the design space pruning techniques described in Section IV are applied incrementally. Each technique is implemented by adding its corresponding constraint step-by-step. Among the techniques, SDP (Structure-Driven Partitioning) can be applied to the 2-bit FF due to its clearly separable dual FF structure with distinct input-output pairs. In

TABLE IV

SO3-Cell generation statistics. OPT refers to the applied optimization methodology. NCT denotes a new circuit topology discovered during layout generation that better fits the given architecture and design rules, defined by comparison with the ASAP7 standard cell library [18].

Cell Specification			Cell Evaluation Metrics										Cell Generation		
Name	#FETs	Height	#CPPs	#M ₁	#M ₂	PAF	CPF	VPT	HPT	V2HVA	H2VVA		OPT	NCT	Time(s)
AOI/OAI21_X1	6	SH	4	0	0	9.69	2.69	5	4	20	28		SO3	✓	3.53
AOI/OAI211_X1	8	SH	5	0	0	10.38	2.67	7	4	28	36		SO3	✓	35.12
AOI/OAI22_X1	8	SH	5	1	0	4.03	2.67	5	4	20	36		SO2-PR	N/A	9.68
AOI/OAI221_X1	10	SH	6	1	0	4.44	2.67	7	4	28	44		SO3	✓	370.45
AOI222/OAI222_X1	12	SH	8	0	0	8.29	2.67	11	4	44	60		SO2-PR	N/A	187.52
		DH	8	2	0	0.00	2.69	3	8	30	70		SO2-PR	N/A	25.83
XOR/XNOR2_X1	10	SH	6	4	1	2.64	2.22	4	3	16	34		SO3	✓	145.45
HA_X1	10	SH	6	4	1	2.90	2.67	4	3	16	33		SO3	✓	364.03
FA_X1	24	SH	13	5	2	3.70	2.67	13	2	52	62		SO2-PR	N/A	1349.30
		DH	14	2	0	30.94	2.67	7	8	71	130		SO2-PR	N/A	2523.89
MUX2_X1	12	SH	7	1	0	5.42	2.22	8	4	32	52		SO2-PR	N/A	17.87
LHQ_X1	16	SH	10	0	0	18.04	2.67	14	4	56	72		SO2-PR	N/A	111.49
DFFHQ_N_X1	24	SH	14	3	1	8.52	2.22	17	3	68	98		SO2-PR	N/A	378.31
		DH	14	4	0	11.08	2.67	5	8	53	130		SO2-PR	N/A	1125.91
2BDFFHQ_N_X1	44	SH	25	8	2	1.57	1.78	28	3	112	173		SO2-PR	N/A	25862.29
		DH	26	2	0	16.44	2.67	16	8	160	240		SO2-PR	N/A	16382.22
DFFRNQ_X1	28	SH	17	7	3	2.52	2.22	17	1	68	76		SO2-PR	N/A	[†] 2853.44
		DH	18	4	0	6.09	2.22	8	8	83	160		SO2-PR	N/A	[†] 25168.95

[†]: DFFRNQ, the most complex cell among the cell set, achieved the reported runtime by constraining the locations of dummy transistors.

TABLE V

The delta in cell metrics between SO3 and PROBE3.0 [13], calculated as (SO3 - PROBE3.0). With the exception of Area, all metrics are higher-is-better. Accordingly, **blue** values represent an improvement over PROBE3.0, while **red** values indicate a degradation.

Cell	Difference (SO3 - PROBE3.0 [13])						
	PAF	CPF	VPT	HPT	V2HVA	H2VVA	Area
AND2_X1	10.46	0.45	3	0	10	0	0
AND3_X1	15.15	0.89	5	0	18	0	0
AND3_X2	22.38	0.45	5	0	19	0	0
AOI21_X1	7.89	0.45	4	0	14	0	0
AOI22_X1	1.28	0.89	4	0	13	0	0
BUF_X1	13.87	0.46	3	0	11	0	0
DFFHQ_N_X1	5.77	1.78	11	3	39	48	-3
DFFRNQ_X1	2.52	2.22	14	1	46	21	-2
INV_X1	3.59	0	2	0	7	0	0
LHQ_X1	16.83	1.78	8	1	29	-2	-1
MUX2_X1	2.54	0.44	5	1	18	5	-1
NAND2_X1	8.4	0.46	3	0	11	0	0
NAND3_X1	11.8	0.45	4	0	13	0	0
NAND4_X1	11.61	0.89	5	0	18	0	0
NOR2_X1	6.47	0.46	4	0	15	0	0
NOR3_X1	10.74	0.45	4	0	14	0	0
NOR4_X1	11.42	0.89	6	0	20	0	0
OAI21_X1	7.26	0.45	4	0	15	0	0
OAI22_X1	1.21	0.89	4	0	14	0	0
OR2_X1	8.8	0.45	3	0	10	0	0
OR3_X1	14.42	0.89	5	0	18	0	0
XOR2_X1	0.22	0.45	1	0	3	-1	0

contrast, for other circuits, the relative positions of transistors are not known in advance, making it infeasible to apply SDP. Additionally, MGS (Misaligned Gate Specification) cannot be applied to FA_X1 because the circuit is fully complementary CMOS logic.

The results demonstrate significant runtime reduction: 89.9% for FA_X1, 92.3% for the 1-bit FF, and 86.1% for the 2-bit FF. The final layout remains unchanged when comparing the baseline (None) with the fully applied set of techniques. This confirms that our pruning techniques do not compromise optimality.

C. Block Experiments

Table VII presents a block PPA comparison and key features of three libraries generated by the SO3-Cell framework, alongside the PROBE3.0 library. The transition from PROBE3.0 to L_1 demonstrates cell generation engine enhancements. The shift from L_1 to L_2 high-

lights improvements in cell scalability and diversity, while the final step from L_2 to L_3 reveals the impact of applying the DH architecture.

We use three RTL designs [20]—aes_cipher_top (AES, 530 FFs), jpeg_encoder (JPEG, 4454 FFs), and vga_enh_top (VGA, 17041 FFs)—to evaluate whether our cell-level improvements consistently lead to block-level improvements across diverse designs. For each RTL, we compare frequency, power, and area. For the block design, the objective is to minimize area while satisfying the following constraints: worst negative slack (WNS) < 20 ps, design rule violations (DRV) < 200, and Shorts < 20. The corresponding power, wire length, and frequency values are collected from this optimized configuration. Frequency is computed as the reciprocal of the effective clock period, defined as the target clock period minus the average timing slack of the 100-worst paths.

The L_1 library achieves an 11.5% improvement in power, a 0.5% improvement in frequency, and a 24.6% area reduction compared to the PROBE3.0 library. This demonstrates the effectiveness of the SO3-Cell framework in enhancing cell layout patterns by incorporating circuit topology optimization and Cell-Flex metrics.

With the expanded cell set in the L_2 library, we observe a 23% improvement in power, a 0.3% improvement in frequency, and a 24.6% reduction in area. The addition of new cells significantly enhances the overall PPA. In particular, the additional power reduction in L_2 is largely attributed to the introduction of 2-bit FF, which considerably reduces dynamic power by minimizing clock delivery network overhead and internal power within the FFs [21]. To analyze the impact of newly added cells, Table VIII presents the area contributions of each cell to the final block area in the SO3-Cell L_2 library across different RTL implementations. Newly introduced combinational cells such as FA_X1, OAI222_X1, AOI222_X1, and XNOR2_X1 rank among the top seven by area contribution. The banking ratio of 2-bit FF reaches approximately 98%, indicating that nearly all FFs were successfully replaced with their 2-bit counterparts.

Transitioning from L_2 to L_3 , we incorporate the DH architecture and replace FA_X1 and FF cells with their DH variants. Although individual DH cells occupy more area, they show notable improvements in cell evaluation metrics, particularly in TUF. L_3 library achieves the best overall PPA improvement, with a 35% reduction in power, a 2.2% increase in frequency, and a 31.1% reduction in

TABLE VI

Simultaneous transistor placement and internal cell routing running time (s) of complex cells when applying design space pruning techniques. (SB: Symmetry Breaking; MOT: Maximum Occupying Track; MGS: Misaligned Gate Specification; SDP: Structure-Driven Partitioning; N/A: not applicable due to circuit structure; X: runtime exceeds 60 hours.)

Cell Specification			Applied Search Sapce Pruning Techniques					Improvement
Name	#FETs	Height	None	SB	SB+MOT	SB+MOT+MGS	SB+MOT+MGS+SDP	
FA_X1	24	SH	13338.87	9067.18	1349.30	N/A	N/A	89.9% ↓
DFFHQ_N_X1	24	SH	4885.50	2919.65	680.60	378.31	N/A	92.3% ↓
2BDDFFHQ_N_X1	44	SH	X	X	X	185942.66	25862.29	86.1% ↓

TABLE VII

Comparison of block-level implementations across four libraries in terms of PPA—power (mW), frequency (GHz), area (μm^2)—and total wire length (μm). PPA values are measured from the smallest block satisfying WNS < 20ps, DRV < 200, and Shorts < 20. L_1 reflects gains from SO3-Cell layout pattern improvement, L_2 from newly added cells, and L_3 from adopting DH architecture. 51* includes 4-to-6-input cells (AOI/OAI{211, 221, 222}_X1), multi-output cells (HA_X1, FA_X1), XNOR_X1 and 2-bit FF (2BDDFFHQ_N_X1).

Library		Metrics	AES		JPEG		VGA		Norm. Avg.	
Name	PROBE3.0 [13]	Power	5.52	0%	7.08	0%	39.46	0%	1.00	0%
		Freq.	7.57	0%	3.05	0%	5.78	0%	1.00	0%
# of cells	41	Area	562.18	0%	2105.97	0%	4039.33	0%	1.00	0%
Architecture	SH	WL	33834.24	0%	75224.45	0%	191667.55	0%	1.00	0%
Name	SO3-Cell L_1	Power	4.90	-11.3%	6.29	-11.2%	34.73	-12.0%	0.88	-11.5%
		Freq.	7.71	+1.9%	3.00	-1.5%	5.86	+1.3%	1.01	+0.5%
# of cells	41	Area	422.35	-24.9%	1496.98	-28.9%	3234.80	-20.1%	0.75	-24.6%
Architecture	SH	WL	29177.92	-11.3%	70504.46	-6.3%	175683.74	-8.3%	0.91	-9.5%
Name	SO3-Cell L_2	Power	4.92	-10.9%	5.14	-27.5%	27.4	-30.6%	0.77	-23.0%
		Freq.	7.34	-2.3%	3.12	+2.5%	5.83	+0.9%	1.00	+0.3%
# of cells	51*	Area	417.55	-25.7%	1304.10	-38.1%	3009.05	-25.7%	0.70	-29.8%
Architecture	SH	WL	29912.25	-11.6%	61620.79	-18.1%	207481.23	+8.3%	0.93	-7.1%
Name	SO3-Cell L_3	Power	4.53	-17.9%	4.36	-38.5%	20.29	-48.6%	0.65	-35.0%
		Freq.	7.74	+2.2%	3.06	+0.2%	6.03	+4.2%	1.02	+2.2%
# of cells	51*	Area	415.72	-26.1%	1295.66	-38.5%	2883.03	-28.8%	0.69	-31.1%
Architecture	SH+DH	WL	28408.72	-16.0%	56523.70	-24.9%	149507.71	-22.0%	0.79	-21.0%

TABLE VIII

Top 7 cell area contributions and 2-bit FF banking ratios across AES, JPEG, and VGA blocks implemented using the SO3-Cell L_2 library. Portion refers to the percentage of block area occupied by each cell type, and Cumulative indicates the total area percentage up to and including that rank. † indicates newly added cells in the SO3-Cell L_2 library.

Rank	AES (FF Banking Ratio: 97.36%)				JPEG (FF Banking Ratio: 98.34%)				VGA (FF Banking Ratio: 98.16%)			
	Cell Name	Portion (%)	Cumulative (%)		Cell Name	Portion (%)	Cumulative (%)		Cell Name	Portion (%)	Cumulative (%)	
1	†2BDDFFHQ_N_X1	11.63	11.63		†2BDDFFHQ_N_X1	29.78	29.78		†2BDDFFHQ_N_X1	52.92	52.92	
2	NAND4_X1	8.54	20.16		†FA_X1	23.37	53.14		OAI22_X1	21.14	74.06	
3	NAND2_X1	7.33	27.49		OAI22_X1	11.69	64.84		AOI22_X1	11.38	85.44	
4	NAND2_X2	5.44	32.94		INV_X1	6.62	71.45		BUF_X1	3.71	89.16	
5	XOR2_X1	5.44	38.38		†OAI222_X1	5.13	76.58		NAND4_X1	2.63	91.79	
6	†XNOR2_X1	4.90	43.28		†AOI222_X1	4.97	81.55		DFFRNQ_X1	1.31	93.10	
7	INV_X1	4.48	47.76		BUF_X1	4.53	86.09		BUF_X2	0.92	94.03	

area. Additionally, a 21% reduction in total wire length helps explain the overall enhancement in block-level PPA.

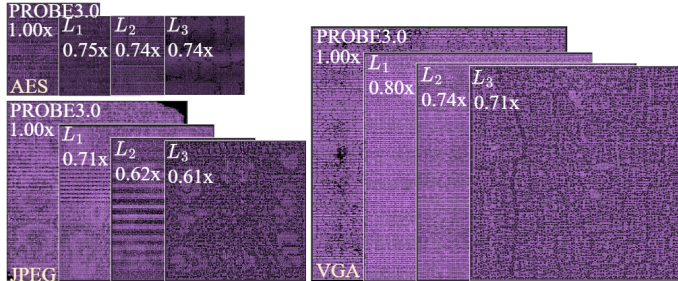


Fig. 13. Block layout snapshots of three different RTL implementations using the PROBE3.0 [13] and SO3-Cell L_1 , L_2 and L_3 libraries. The SO3-Cell L_1 library, which integrates circuit topology optimization and Cell-Flex metrics, achieves a normalized area of 0.75 for AES, 0.71 for JPEG, and 0.75 for VGA. Additional gains can be obtained by extending the cell set and applying the Double-Height (DH) architecture.

VII. CONCLUSION

This paper presents SO3-Cell, the first standard cell layout automation framework that simultaneously optimizes circuit topology, transistor placement, and internal cell routing within a unified framework under a single objective function. To overcome the limitations of heuristic placement algorithms, we introduce several design space pruning techniques that significantly mitigate complexity, reducing the runtime of complex cells by up to 90%. We successfully generate a layout for a 44-transistor 2-bit flip-flop within 25,862 seconds, demonstrating the scalability and robustness of the SO3-Cell framework. Our MILP-based formulation incorporates constraints to enhance *cell layout flexibility* based on the Cell-Flex metrics, effectively improving the layout patterns generated by SO3-Cell. Through block-level experiments, we validate the effectiveness of the proposed framework, demonstrating improvements in cell scalability and additional gains from applying DH architectures. Compared with the existing PROBE3.0 [13] library, SO3-Cell achieves significant PPA improvements, including a 35.0% reduction in power, a 2.2% increase in frequency, and a 31.1% reduction in area, confirming the practical impact of our framework.

ACKNOWLEDGEMENT

This work is supported by NSF under Grant CCF-2110419.

REFERENCES

- [1] C.-K. Cheng, B. Kang, B. Lin, and Y. Wang, "Scaling Standard Cell Layout Using Track Height Compression and Design Technology Co-optimization," in *Proceedings of the International Symposium on Physical Design*. ACM, 2025, pp. 11–19.
- [2] K. Jo, S. Ahn, J. Do, T. Song, T. Kim, and K. Choi, "Design Rule Evaluation Framework Using Automatic Cell Layout Generator for Design Technology Co-Optimization," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 27, no. 8, pp. 1933–1946, 2019.
- [3] S. Chakravarty, X. He, and S. Ravi, "Minimum area layout of series-parallel transistor networks is NP-hard," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 10, no. 7, pp. 943–949, 1991.
- [4] C.-K. Cheng, B. Kang, B. Lin, and Y. Wang, "Standard Cell Layout Generation: Review, Challenges, and Future Works," in *Proceedings of the Asia and South Pacific Design Automation Conference*. ACM, 2025, pp. 372–378.
- [5] Y.-L. Li, S.-T. Lin, S. Nishizawa, and H. Onodera, "Mcell: Multi-row cell layout synthesis with resource constrained MAX-SAT based detailed routing," in *Proceedings of the International Conference on Computer-Aided Design*. ACM, 2020, pp. 1–8.
- [6] K. Baek and T. Kim, "Csyn-fp: standard cell synthesis of advanced nodes with simultaneous transistor folding and placement," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 43, no. 2, pp. 627–640, 2023.
- [7] Y.-L. Li, S.-T. Lin, S. Nishizawa, H.-Y. Su, M.-J. Fong, O. Chen, and H. Onodera, "NCTUcell: A DDA-aware cell library generator for FinFET structure with implicitly adjustable grid map," in *Proceedings of the Design Automation Conference*. ACM, 2019, pp. 1–6.
- [8] J.-C. Tsai, W.-M. Hsu, Y.-T. Hsieh, Y.-J. Li, W. Huang, C. Ho, H.-M. Huang, J.-H. Yang, H.-L. Huang, A. C.-W. Liang *et al.*, "MAXCell: PPA-Directed Multi-Height Cell Layout Routing Optimization using Anytime MaXSAT with Constraint Learning," in *Proceedings of the International Conference on Computer-Aided Design*. ACM, 2024, pp. 1–9.
- [9] D. Lee, D. Park, C.-T. Ho, I. Kang, H. Kim, S. Gao, B. Lin, and C.-K. Cheng, "SP&R: SMT-Based Simultaneous Place-and-Route for Standard Cell Synthesis of Advanced Nodes," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 40, no. 10, pp. 2142–2155, 2021.
- [10] H. Ren and M. Fojtik, "Nvcell: Standard cell layout in advanced technology nodes with reinforcement learning," in *Proceedings of the Design Automation Conference*. IEEE, 2021, pp. 1291–1294.
- [11] C.-K. Cheng, D. Lee, and D. Park, "Standard-Cell Scaling Framework with Guaranteed Pin-Accessibility," in *Proceedings of the International Symposium on Circuits and Systems*. IEEE, 2020, pp. 1–5.
- [12] B. Kang, Y. Yuan, Y. Wang, B. Lin, and C.-K. Cheng, "Cell-Flex Metrics for Designing Optimal Standard Cell Layout with Enhanced Cell Layout Flexibility," in *Proceedings of the International Symposium on Physical Design*. ACM, 2025, pp. 164–172.
- [13] S. Choi, J. Jung, A. B. Kahng, M. Kim, C.-H. Park, B. Pramanik, and D. Yoon, "PROBE3.0: A Systematic Framework for Design-Technology Pathfinding With Improved Design Enablement," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 43, no. 4, pp. 1218–1231, 2023.
- [14] S. Y. Sherazi, M. Cupak, P. Weckx, O. Zografos, D. Jang, P. Debacker, D. Verkest, A. Mocuta, R. Kim, A. Spessot *et al.*, "Standard-cell design architecture options below 5nm node: The ultimate scaling of FinFET and Nanosheet," in *Proceedings of the Design-Process-Technology Co-optimization for Manufacturability XIII*, vol. 10962. SPIE, 2019, p. 1096202.
- [15] K. Han, A. B. Kahng, and H. Lee, "Evaluation of BEOL design rule impacts using an optimal ILP-based detailed router," in *Proceedings of the Design Automation Conference*. ACM, 2015, pp. 1–6.
- [16] N. Ryzhenko and S. Burns, "Standard cell routing via boolean satisfiability," in *Proceedings of the Design Automation Conference*. ACM, 2012, pp. 603–612.
- [17] S. Kim, "Optimal Power Delivery Strategy in Modern VLSI Design," Ph.D. dissertation, Graduate School of UNIST, 2019.
- [18] L. T. Clark, V. Vashishtha, L. Shifren, A. Gujja, S. Sinha, B. Cline, C. Ramamurthy, and G. Yeric, "ASAP7: A 7-nm finFET predictive process design kit," *Microelectronics Journal*, vol. 53, pp. 105–115, 2016.
- [19] "SO3-Cell," GitHub repository. [Online]. Available: <https://github.com/ckchengucsd/SO3-Cell>
- [20] OpenCores, "Open-source IP cores." [Online]. Available: <https://opencores.org/>
- [21] C. Santos, R. Reis, G. Godoi, M. Barros, and F. Duarte, "Multi-bit flip-flop usage impact on physical synthesis," in *Proceedings of the Symposium on Integrated Circuits and Systems Design*. IEEE, 2012, pp. 1–6.